

BANGLADESH UNIVERSITY OF PROFESSIONALS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Digital Communication

Bandwidth, power efficiency, Modulation and coding trade-off, Digital band pass transmission

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Contents

- ❑ Bandwidth, Power Efficiency
- ❑ Modulation and coding trade-off
- ❑ Digital band pass transmission

Bandwidth

❑ Bandwidth

A bandwidth indicates the maximum amount of data that can be transmitted or received simultaneously over the communication channel.

- Units: Hertz(Hz), kHz, MHz, GHz, Kbps, Mbps, Gbps
- Example:
  1. FM Radio uses the frequency range of 88-108 MHz.
  2. Internet speed like 15 Gbps.
- ***Analog Bandwidth:*** Range of frequencies that an analog signal occupies. An Analog signal from 300 Hz to 3.4 kHz has 3.1 kHz Bandwidth.
- ***Digital Bandwidth:*** Amount of data that can be transmitted per second. Internet speed like 15 Gbps (Giga bits per second).

Bandwidth

❑ Bandwidth

A bandwidth indicates the maximum amount of data that can be transmitted or received simultaneously over the communication channel.

- ***Baseband Bandwidth:*** Bandwidth of a signal before modulation (original frequency starting from 0 Hz). Use-case: Ethernet.
- ***Passband Bandwidth:*** Bandwidth of a modulated signal shifted to a higher frequency range (carrier-based transmission). Use-case: : Wireless Communication, RF systems.
- ***Broadband Bandwidth:*** High-capacity bandwidth allowing multiple signals or channels over a medium. Use-case: Cable TV, Fibre Optics.

Bandwidth Efficiency

❑ Bandwidth & Bandwidth Efficiency

- ***Bandwidth:*** The physical range of frequencies (in Hz) occupied by a signal. It is a finite, expensive resource.

- ***Bandwidth Efficiency:*** Also called Spectral Efficiency, it measures how many bits of information you can cram into each Hertz of that bandwidth.

$$Bandwidth\ Efficiecy, \eta = \frac{R}{B}\ bps/Hz$$

R = Information Rate/ Digital Bandwidth

B = Bandwidth

Bandwidth Efficiency

❑ Bandwidth Efficiency

$$Bandwidth\ Efficiecy, \eta = \frac{R}{B}\ bps/Hz$$

- ***Low Bandwidth:*** You have a wide Analog pipe (high B) but only send a few bits per second (low R).

- ***High Efficiency:*** You have a very narrow Analog pipe (low B) but use advanced modulation to cram a massive bit rate (high R) through it.

Power

Power of a signal can be calculated from the 2nd moment:

$$E\{x^2\} = \int_{-\infty}^{+\infty} x^2 \, pdf(x) \, dx$$

Power Efficiency

Power efficiency tells us how effectively a communication system uses transmitted power to achieve reliable data transmission.
Power efficiency is typically measured by the required ***Energy per Bit (E_b)*** relative to the ***Noise Power Spectral Density (N_0)*** to achieve a target error probability.

Power efficiency can be defined by-

$$Power\ Efficiency\ is\ evaluated\ as \frac{E_b}{N_0}$$

Here, E_b = Energy per bit,
N_0 = Noise Power Spectral Density

Power Efficiency

Power efficiency can be evaluated with-

$$Power\ Efficiency\ is\ evaluated\ as\ \frac{E_b}{N_0}$$

Here, E_b = Energy per bit,

N_0 = Noise Power Spectral Density and

$$Energy\ per\ bit, E_b = \frac{P}{R}$$

Here, P = Total energy of a signal

R = Bit Rate (Digital Bandwidth)

❑ Power Efficiency

Power efficiency can be evaluated with-

$$Power\ Efficiency\ is\ evaluated\ as\ \frac{E_b}{N_0}$$

Here, E_b = Energy per bit,

N_0 = Noise Power Spectral Density and

- To calculate Noise Power Spectral Density (N_0), we look at the physical nature of noise in a communication channel: Thermal Noise
- Modeled as a "White Noise" because its power is spread uniformly across the frequency spectrum.

Power Efficiency

Power efficiency can be evaluated with-

$$Power\ Efficiency\ is\ evaluated\ as\ \frac{E_b}{N_0}$$

Here, E_b = Energy per bit,

N_0 = Noise Power Spectral Density and

$$N_0 = k \times T$$

Here, k = Boltzmann's constant $\approx 1.38 \times 10^{-23}$ Joules per Kelvin (J/K).

T = The absolute temperature of the receiver/system in Kelvin (K).

- **Power Efficiency vs Bandwidth Efficiency**

The formulas for representing the Power Efficiency and Bandwidth Efficiency-

$$Power\ Efficiency\ is\ evaluated\ as\ \frac{E_b}{N_0}$$

$$\Rightarrow Power\ Efficiency\ is\ evaluated\ as\ \frac{P}{R}\times\frac{1}{N_0}\ldots\ldots\ldots\ldots\ldots(i)[Since, E_b=\frac{P}{R}]$$

$$Bandwidth\ Efficiecy, \eta=\frac{R}{B}bps/Hz\ldots\ldots\ldots\ldots\ldots(ii)$$

Considering the common term, R, we can imply, the Power Efficiency and the Bandwidth Efficiency are reverse proportional.

Power Efficiency vs Bandwidth Efficiency

Signal-to-Noise-Ratio (SNR): SNR is a measure of the strength of the desired information-bearing signal relative to the unwanted background noise.

- It determines the clarity of the received signal. We calculate SNR with:

$$\boldsymbol{SNR = \frac{P_{signal}}{P_{noise}} = \frac{P}{N}}$$

$$\boldsymbol{SNR_{dB} = 10\log_{10}\frac{P_{signal}}{P_{noise}}}$$

Where, P_{signal} = average power of the signal

P_{Noise} = average power of the noise

High SNR: The signal is much stronger than the noise

Low SNR: The signal is weaker than the noise. The signal is buried in noise.

Power Efficiency & Bandwidth Efficiency

Power Efficiency vs Bandwidth Efficiency

Channel Capacity (C): the theoretical upper bound on the rate at which information can be transmitted reliably (with an arbitrarily small error probability) over a communication channel.

- It is essentially the "speed limit" of the physics of the medium.

The Capacity of a Gaussian Channel:

$$\boldsymbol{C = B \log_2(1 + SNR)}$$

Where, C = Capacity in bits per second

B = Analog Bandwidth

SNR = Signal-to-Noise-Ratio in linear scale

Power Efficiency & Bandwidth Efficiency

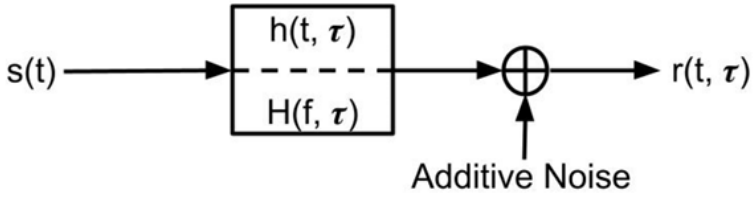
- **Power Efficiency vs Bandwidth Efficiency**

  See Appendix 6.1

Mid Question Overview

Question-1 a.	A sensor in an industrial automation system produces a continuous voltage signal X. The Cumulative Distribution Function (CDF) of the signal is modeled by the following logistic growth function. $$CDF_X(x) = \frac{1}{1+e^{-x}}$$ **Evaluate** the probability that the amplitude of the signal at any given time will be in between 1 and 5 Volts inclusive. Show your calculation steps.	4
b.	A stochastic signal $X(t)$ represents the output of a high-frequency sensor. The statistical behavior of the signal amplitude is described by the following Probability Density Function (PDF). $$PDF_X(x) = \begin{cases} 5e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$ **Calculate** the total average power of the signal.	3
c.	**Demonstrate** whether the PDF given in Question-1(b) satisfies the Exponential Distribution or not.	3

Mid Question Overview

Question-2 a.	Explain the advantage of a Digital Communication System over an Analog Communication System from the receiver's perspective.	2
b.	Describe the responsibilities of a transmitter in a Digital Communication System in detail.	4
c.	Discuss the properties of the filter used in the following channel. Explain the distortions that the channel introduces. 	4

Mid Question Overview

Question-3 a.	At any given time, the transmitter of a Digital Communication System receives the following string from its information source. W W W X # X # # # K K K L L W W K % L W Find out the Huffman codes for the symbols. Ignore the spaces in between.	4
b.	Define Information and Entropy from the perspective of Digital Communication. Calculate the Information and Entropy from the Huffman codes found in Question-3(a).	3
c.	Explain efficiency of a source coding scheme. Calculate the efficiency for the Huffman coding found in Question-3(a)	3

Mid Question Overview

Question-4 a.	Explain the causes of inter-symbol interference.	2
b.	Discuss the encoding and decoding procedures of a Linear Predictive Coding model.	5
c.	Describe the characteristics and applications of a Baseband Digital Transmission System.	3

Your Feedback

1. How would you rate the pace of the lectures so far?
2. Which specific topic from the Midterm syllabus did you find the most difficult to grasp?
3. On a scale of 1-5, how effectively do the mathematical derivations in class translate to your understanding of the conceptual "big picture"?
4. Are the provided lecture slides and notes sufficient for your self-study?
5. Did the Midterm exam accurately reflects the topics covered in the lectures